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Bollinger Bands Initial Backtest Report

The Bollinger Bands trading strategy is a simple one that is based around mean reversion. It was invented in the 1980's by John Bollinger. The strategy has three parameters, EntryZscore, ExitZscore, and rollingWindow. The strategy dictates that you enter a position if the price deviates by more than EntryZscore standard deviations from the mean. If the price crosses ExitZscore standard deviations from the mean, then you exit the position. The mean and standard deviations are computed based on the rollingWindow. At any one time the strategy has zero or one units invested and can go both long and short.

I discovered this strategy in a book entitled 'Algorithmic Trading' by Ernest P Chan. Chan tests out the strategy on the spread GLD-USO spread, but notes that the strategy should still be applicable using ratios instead of spreads. Chan arbitrarily set his EntryZscore to 1, ExitZscore to 0, and rollingWindow to 20, noting that when conducting a thorough backtest, these values should be optimized. He ran the strategy from May 24th 2006 until April 9th 2012. It generated an APR of 17.8% and a sharpe ratio of 0.96. The equity curve is just to the right.

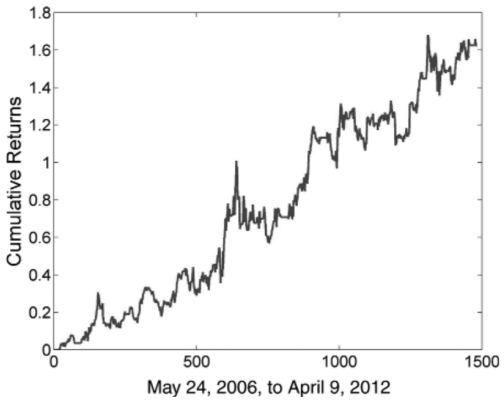

FIGURE 3.3 Cumulative Returns of Bollinger Band Strategy on GLD-USO

For my first backtest, I implemented the same conditions as Chan did, setting my entry and exit z-scores to 1 and 0 respectively, and my rollingWindow to 20. I tested this strategy on 5 different currencies, EUR-GBP, AUD-USD, and NZD-USD. The performance and graphs for each are attached below

<u>EUR-GBP</u>

```
Start                     1999-01-04 00:00:00
End                       2024-02-02 00:00:00
Duration                   9160 days 00:00:00
Exposure Time [%]                  83.399633
Equity Final [$]                    4936.468
Equity Peak [$]                 10353.248738
Return [%]                         -50.63532
Buy & Hold Return [%]              19.981715
Return (Ann.) [%]                  -2.685075
Volatility (Ann.) [%]               7.049496
Sharpe Ratio                             0.0
Sortino Ratio                            0.0
Calmar Ratio                             0.0
Max. Drawdown [%]                 -52.971855
Avg. Drawdown [%]                  -8.522673
Max. Drawdown Duration      8421 days 00:00:00
Avg. Drawdown Duration      1141 days 00:00:00
# Trades                                 841
Win Rate [%]                       44.827586
Best Trade [%]                      8.631405
Worst Trade [%]                    -6.972136
Avg. Trade [%]                      -0.08391
Max. Trade Duration           35 days 00:00:00
Avg. Trade Duration            9 days 00:00:00
Profit Factor                       0.828543
Expectancy [%]                     -0.077339
SQN                                -1.891548
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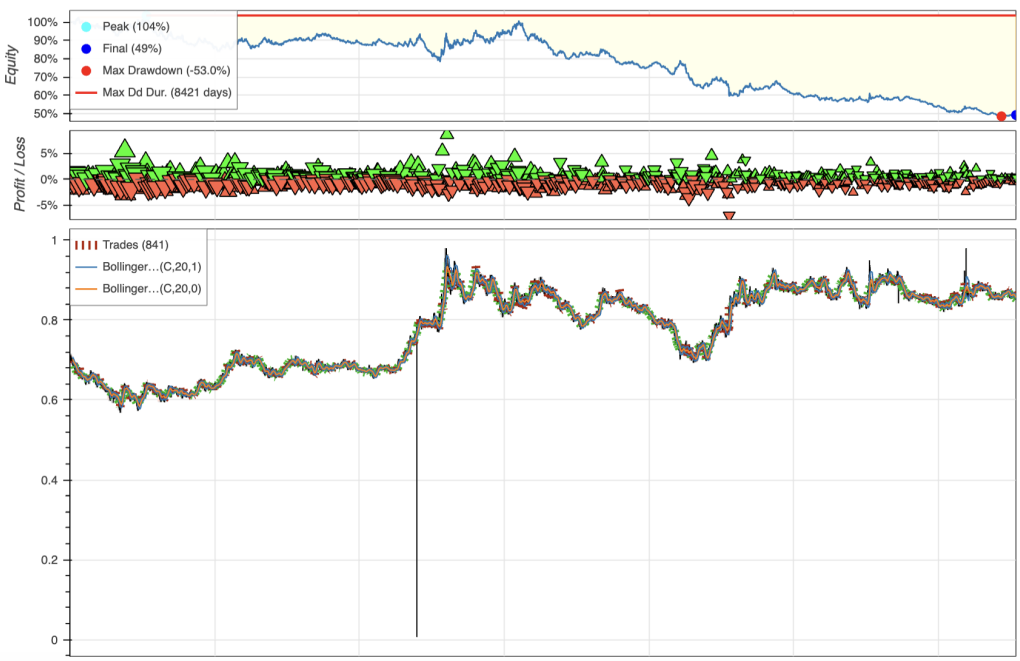

AUD-USD

Start	2008-04-21 00:00:00
End	2024-02-02 00:00:00
Duration	5765 days 00:00:00
Exposure Time [%]	86.357004
Equity Final [$]	12421.689624
Equity Peak [$]	14660.703054
Return [%]	24.216896
Buy & Hold Return [%]	-30.31265
Return (Ann.) [%]	1.33787
Volatility (Ann.) [%]	11.370685
Sharpe Ratio	0.11766
Sortino Ratio	0.17143
Calmar Ratio	0.045561
Max. Drawdown [%]	-29.364626
Avg. Drawdown [%]	-4.754128
Max. Drawdown Duration	4946 days 00:00:00
Avg. Drawdown Duration	285 days 00:00:00
# Trades	525
Win Rate [%]	48.0
Best Trade [%]	22.137969
Worst Trade [%]	-9.037587
Avg. Trade [%]	0.041316
Max. Trade Duration	48 days 00:00:00
Avg. Trade Duration	9 days 00:00:00
Profit Factor	1.100973
Expectancy [%]	0.064704
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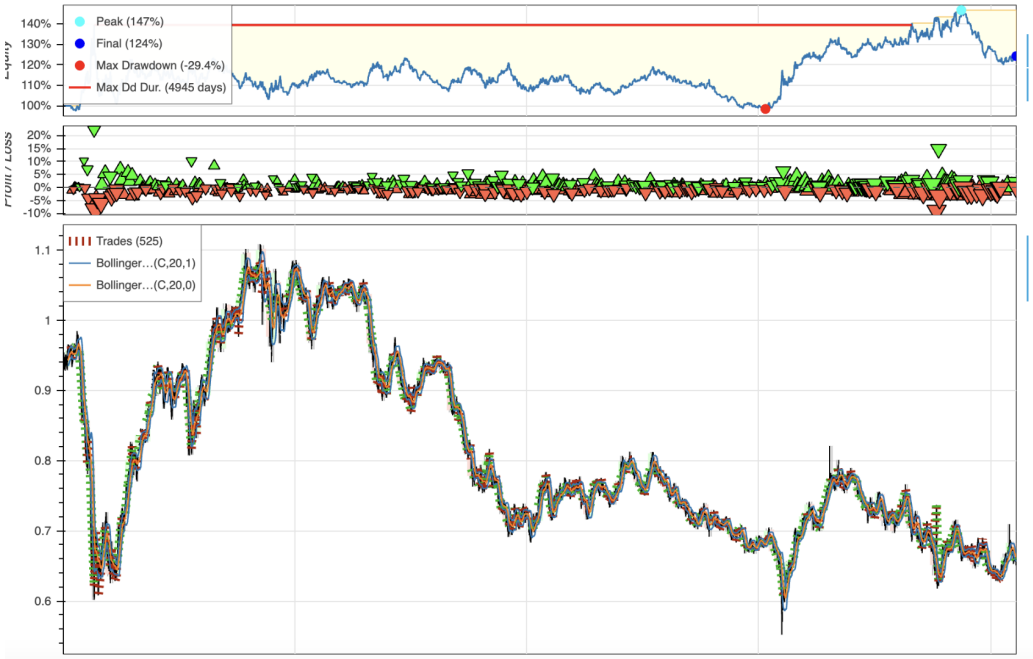

NZD-USD

Start	2003-12-01 00:00:00
End	2024-02-02 00:00:00
Duration	7368 days 00:00:00
Exposure Time [%]	85.736108
Equity Final [$]	10694.296791
Equity Peak [$]	13101.303796
Return [%]	6.942968
Buy & Hold Return [%]	-4.61237
Return (Ann.) [%]	0.323524
Volatility (Ann.) [%]	11.266622
Sharpe Ratio	0.028715
Sortino Ratio	0.041664
Calmar Ratio	0.010039
Max. Drawdown [%]	-32.227807
Avg. Drawdown [%]	-7.314184
Max. Drawdown Duration	4230 days 00:00:00
Avg. Drawdown Duration	404 days 00:00:00
# Trades	646
Win Rate [%]	45.975232
Best Trade [%]	12.125097
Worst Trade [%]	-4.55399
Avg. Trade [%]	0.010391
Max. Trade Duration	48 days 00:00:00
Avg. Trade Duration	10 days 00:00:00
Profit Factor	1.042051
Expectancy [%]	0.028608
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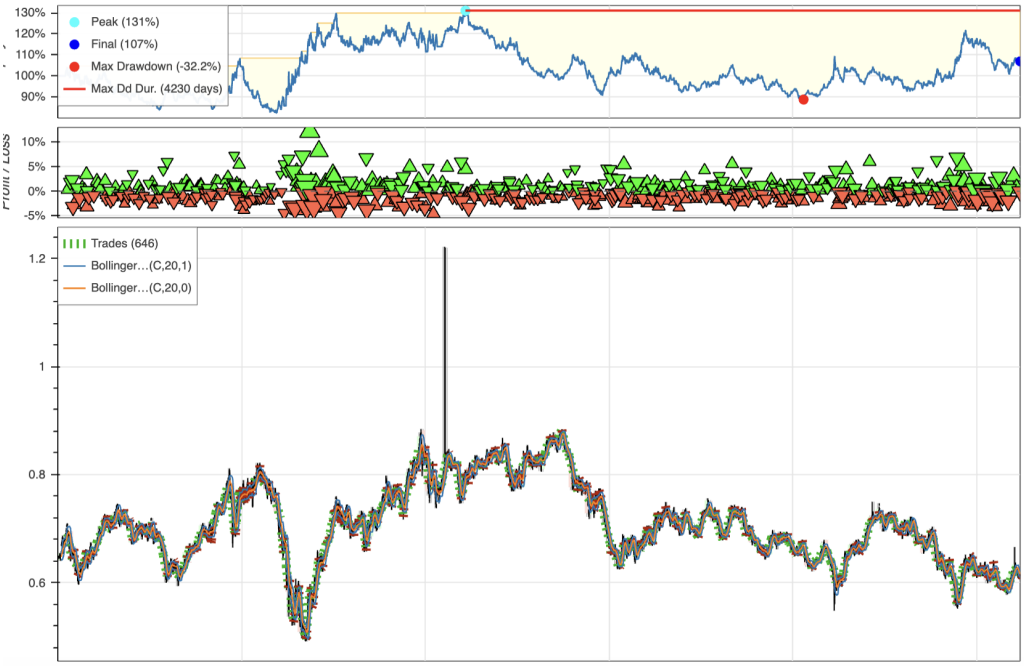

It is evident that in my currency tests, the given strategy doesn't perform as well as it did in the demo, especially given the length of time it was operating for . One reason for this might be slippage. Since this is using end of day data, the buy/sell signals are being computed on the close, but the orders are only going through on the next day open. The strategy did tend to lose money on extremely large movements. I personally believe that the strategy will perform much better once I query more granular data and optimize the strategy.

I have not made any unusual assumptions in my model.

References

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https://kernc.github.io/backtesting.py/doc/backtesting/backtesting.html#backtesting.backtesting.Backtest.optimize

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